

ARCHITECTURE

- Control plane:** kube-apiserver (only door to etcd), etcd, kube-scheduler, kube-controller-manager, cloud-controller-manager.
- Node:** kubelet, kube-proxy (nftables GA 1.33; IPVS deprecated 1.35), container runtime via CRI (containerd / CRI-O; no dockershim since 1.24).
- Declarative: desired state in spec, controllers reconcile to status in a loop.
- Add-ons: CoreDNS, CNI plugin, metrics-server.
- Flow: kubectl → API server (authn → authz → admission) → etcd → scheduler binds → kubelet runs.

LOCAL CLUSTERS

```
kind create cluster --name dev \
  --image kindest/node:v1.37.0
kind load docker-image app:1.0 minikube start --kubernetes-versions=v1.37.1
minikube addons enable ingress
minikube tunnel # LoadBalancer
kubectl config get-contexts
kubectl config use-context kind-dev
kubectl config set-context --current \
  --namespace=bookshelf

Also Docker Desktop, k3d, Kubeconfig = clusters + users + contexts (SKUBECONFIG).
```

KUBECTL

```
kubectl get pods -A -o wide
kubectl get po -l app=web -w
kubectl get deploy web -o yaml
kubectl get po -o jsonpath='{.items[*].metadata.name}'
kubectl get po -o custom-columns=N:.metadata.name
kubectl describe pod web-0
kubectl apply -f app.yaml # declarative
kubectl apply -k overlays/prod
kubectl diff -f app.yaml
kubectl apply --dry-run=server -f app.yaml
kubectl create deployment web --image=nginx:1.29
kubectl delete -f app.yaml
kubectl explain pod.spec.containers
kubectl api-resources; kubectl api-versions
kubectl logs web-0 -c app -f --previous
kubectl exec -it web-0 -- sh
kubectl port-forward svc/web 8080:80
kubectl cp web-0:/tmp/a ./a
kubectl top pod; kubectl top node
kubectl auth can-i list pods --as=su
kubectl get events --sort-by=.lastTimestamp

Selectors -l k=v,k2 in (a,b), --field-selector; -n/-A; kubectl run makes only a Pod; needs -- before exec cmd.
```

OBJECT SKELETON

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: web
  namespace: bookshelf
  labels:
    app.kubernetes.io/name: web
    annotations: {note: "free-for m"}
spec: {...} # desired state
# status: written by controllers
```

- Labels select (selector); annotations don't. Names unique per kind+namespace; uid per object.
- ownerReferences + finalizers drive garbage collection.
- Server-side apply: --server-side, field managers.

WORKLOADS + ROLLOUT

Kind	Use
Pod	smallest unit; shared net/volumes; native sidecars GA 1.33
Deployment	stateless; owns ReplicaSets; rolling update
StatefulSet	stable names/storage, ordered; headless Svc
DaemonSet	one Pod per node
Job / CronJob	run-to-completion / scheduled (batch/v1)

```
kubectl rollout status deploy/web
kubectl rollout history deploy/web
kubectl rollout undo deploy/web --to-revision=2
kubectl rollout restart deploy/web
kubectl rollout pause|resume deploy/web
kubectl scale deploy/web --replicas=5
kubectl set image deploy/web app=img:1.1
strategy: {type: RollingUpdate, rollingUpdate: {maxSurge: 25%, maxUnavailable: 0}}
```

Also Recreate, blue/green, canary. Job: backoffLimit, ttlSecondsAfterFinished; CronJob concurrencyPolicy.

PROBES + LIFECYCLE

```
startupProbe:
  httpGet: {path: /healthz, port: 8080}
failureThreshold: 30
periodSeconds: 5
livenessProbe: {tcpSocket: {port: 8080}}
readinessProbe:
  httpGet: {path: /ready, port: 8080}
periodSeconds: 5
lifecycle:
  preStop: {sleep: {seconds: 5}}
terminationGracePeriodSeconds: 30
```

- Startup gates the others; liveness restarts; readiness removes from Service endpoints.
- Kinds: httpGet, tcpSocket, exec, grpc. Keep liveness cheap, no dependency checks.
- Shutdown: SIGTERM → grace period → SIGKILL.

SERVICES · DNS

Type	Behaviour
ClusterIP	internal virtual IP (default)
NodePort	port 30000-32767 on every node
LoadBalancer	cloud LB in front of NodePort
ExternalName	CNAME to external DNS
Headless	clusterIP: None; DNS returns Pod IPs

```
ports: [{port: 80, targetPort: 8080}]
selector: {app: web}
```

- DNS: svc.ns.svc.cluster.local; StatefulSet Pod: web-0.svc.ns.svc.; SRV for named ports.
- EndpointSlices replace Endpoints (deprecated 1.33); externalIPs deprecated 1.36.

INGRESS VS GATEWAY API

	Ingress	Gateway
Kinds	Ingress + controller	GatewayClass, Gateway, HTTPRoute ...
Roles	one object	infra / cluster op / app dev split
Extras	controller annotations	headers, weights, mirroring, typed
Status	GA, frozen	GA (v1.6), the successor

```
kind: Ingress
spec:
  ingressClassName: nginx
  rules:
  - {host: b.example.com, http: {paths: [{path: /, pathType: Prefix, backend: {service: {name: web, port: {number: 80}}}}]}}
  tls: [{hosts: [b.example.com], secretName: b-tls}]

Ingress-NGINX retired (Mar 2026): pick another controller or migrate to Gateway (ingress2gateway). TLS via cert-manager.
```

NETWORKPOLICY

```
kind: NetworkPolicy
spec:
  podSelector: {}
  policyTypes: [Ingress, Egress]
  # empty rules = default deny

ingress:
  - from: [{podSelector: {matchLabels: {app: web}}}
    ports: [{port: 5432}]
egress:
  - to: [{namespaceSelector: {}}]
    ports: [{port: 53, protocol: UDP}]
```

- Additive allow-lists; no deny rules. Needs a CNI that enforces it (Calico, Cilium; not plain kindnet/flannel).

CONFIGMAP · SECRET

```
kubectl create configmap app-cfg \
  --from-literal=LOG=info --from-file=a.properties
kubectl create secret generic db \
  --from-literal=password='CHANGE_ME'
envFrom: [{configMapRef: {name: app-cfg}}]
env: [{name: PW, valueFrom: {secretKeyRef: {name: db, key: password}}}]
volumes: [{name: c, configMap: {name: app-cfg}}]
```

- Secrets are base64, not encrypted: enable etcd encryption at rest, RBAC, external secrets/CSI driver.
- Volume mounts refresh; env vars don't immutable: true.
- Types: Opaque, kubernetes.io/tls, dockerconfigjson.

STORAGE

Object	Role
PV	cluster storage unit
PVC	namespaced request, bound to PV
StorageClass	dynamic provisioner (CSI)

```
accessModes: [ReadWriteOnce] # RWO, ROX, RWX, RWOP
storageClassName: standard
resources: {requests: {storage: 10Gi}}
volumeBindingMode: WaitForFirstConsumer
reclaimPolicy: Delete|Retain
allowVolumeExpansion: true
```

- Volumes: emptyDir, configMap, secret, projected, hostPath (avoid), generic ephemeral.
- VolumeSnapshots for backup/clone; StatefulSet volumeClaimTemplates.
- In-tree cloud plugins gone (CSI migration).

RESOURCES · QOS

```
resources:
  requests: {cpu: 250m, memory: 256Mi}
  limits: {memory: 512Mi}
```

- Requests drive scheduling; limits enforced by cgroups: CPU throttled, memory OOMKilled (exit 137).
- QoS: Guaranteed (req = limit, all containers) · Burstable · BestEffort: evicted first.
- LimitRange (defaults), ResourceQuota (per namespace).
- In-place Pod resize GA 1.35 (resizePolicy).

SCHEDULING · PDB · DRAIN

```
nodeSelector: {disktype: ssd}
affinity: {nodeAffinity: {podAntiAffinity: ...}
tolerations: [{key: gpu, operator: Exists, effect: NoSchedule}]
topologySpreadConstraints:
  - maxSkew: 1
    topologyKey: topology.kubernetes.io/zone
    whenUnsatisfiable: DoNotSchedule
labelSelector: {...}
priorityClassName: high
kind: PodDisruptionBudget
spec: {minAvailable: 2, selector: {...}}
kubectl cordon|uncordon NODE
kubectl drain NODE --ignore-daemonsets \
  --delete-emptydir-data
kubectl taint nodes N k=NoSchedule
```

PDB guards voluntary disruptions (drain, upgrades) only.

AUTOSCALING

```
kubectl autoscale deploy web \
  --min=2 --max=10 --cpu=70%
apiVersion: autoscaling/v2
kind: HorizontalPodAutoscaler
metrics: [{type: Resource, resource: {name: cpu, target: {type: Utilization, averageUtilization: 70}}}]
behavior: {scaleDown: {stabilizationWindowSeconds: 300}}
```

- HPA needs metrics-server + requests; scale-to-zero beta 1.37.
- VPA (add-on) right-sizes; Cluster Autoscaler / Karpenter add nodes; KEDA for events.

SECURITY

```
kind: Role
# namespaced; ClusterRole = cluster-wide
rules: [{apiGroups: [""], resources: [pods], verbs: [get, list, watch]}]
kubectl create rolebinding r --role=R \
  --serviceaccount=ns:sa
securityContext:
  runAsNonRoot: true
  allowPrivilegeEscalation: false
  readOnlyRootFilesystem: true
  capabilities: {drop: [ALL]}
  seccompProfile: {type: RuntimeDefault}
kubectl label ns app \
  pod-security.kubernetes.io/enforce=restricted \
  pod-security.kubernetes.io/warn=restricted \
  pod-security.kubernetes.io/audit=restricted
```

- RBAC is additive allow-only; avoid cluster-admin and wildcards.
- ServiceAccount tokens: projected, bound, expiring (TokenRequest); no auto-created Secrets since 1.24; automountServiceAccountToken: false.
- PSS: privileged / baseline / restricted (PSP removed 1.25). hostUsers: false (GA 1.36).
- Admission: ValidatingAdmissionPolicy (CEL), Kyverno, Gatekeeper.

KUSTOMIZE

```
resources: [.././base]
namePrefix: prod-
images: [{name: web, newTag: "1.2.0"}]
patches: [{path: replicas.yaml}]
configMapGenerator: [{name: cfg, literals: [LOG=warn]}]
kubectl kustomize overlays/prod
kubectl apply -k overlays/prod
```

Base + overlays; template-free; built into kubectl (5.8).

HELM

```
helm repo add bitnami https://charts.bitnami.com/bitnami
helm install rel ./chart -f values.yaml --set a=b
helm upgrade --install rel oci://reg/chart --version 1.2.3
helm list; helm history rel
helm rollback rel 1
helm template|lint ./chart
helm get values rel; helm show values CHART
helm uninstall rel
```

Chart + values = release. Helm 4: OCI first-class, server-side apply, new plugin system. Pin chart versions; verify provenance.

CRDS · OPERATORS

- CRD adds a kind with OpenAPI schema, versions, validation (CEL x-kubernetes-validations); CR = an instance.
- Operator = CRD + controller reconciling to desired state (Kubebuilder, Operator SDK, controller-runtime).
- kubectl get crd; kubectl explain CR.spec; delete CRD deletes all CRs.

TROUBLESHOOTING

Symptom	Check / fix
Pending	describe pod events: requests too big, taints, unbound PVC
ImagePullBackOff	image name/tag, imagePullSecrets, registry reach
CrashLoopBackOff	logs --previous, probes, config, exit code
OOMKilled (137)	raise memory limit; check leaks
No endpoints	selector vs labels; readiness failing
Svc timeout	get endpointslices; NetworkPolicy; DNS
Evicted	node pressure; set requests
Node NotReady	describe node; kubelet, CNI, disk
kubectl debug -it web-0 --image=busybox \ --target=app kubectl debug node/N -it --image=busybox kubectl run t --rm -it --image=busybox -- \ nslookup web.bookshelf	

CHANGED SINCE OLDER TUTORIALS

- Dockershim removed 1.24 → containerd/CRI-O.
- PSP removed 1.25 → Pod Security Admission.
- No auto token Secrets since 1.24 → TokenRequest.
- extensions/v1beta1 & apps/v1beta1 gone (1.16) → apps/v1; HPA autoscaling/v2.
- kubectl run = Pod only; rolling-update removed 1.18.
- Ingress-NGINX retired Mar 2026 → Gateway API.
- Endpoints → EndpointSlice; IPVS deprecated; cgroup v1 deprecated 1.35.
- Legacy apt/yum.kubernetes.io repos → pkgs.k8s.io.
- In-tree cloud volumes → CSI.
- New: native sidecars GA 1.33, in-place resize GA 1.35, user namespaces GA 1.36.